Intro to Data Science

Lec 4

Regression

- Type of supervised learning
- Regression analysis is a form of predictive modeling technique which investigate the relationship between dependent and independent variable.

Predictive Model

Parametric Model

Non Parametric Model

Parametric Model	**Non-Parametric Model**
that require specification of some parameters before they can used to make prediction.	do not rely on any specific parameter setting.
Algorithm are • linear regression • logistic regression	Algorithm are • Decision tree • Support Vector Machine

Types of Regression

- Linear Regression
- Logistic Regression
- Ridge Regression
- Lasso Regression
- Polynimoal Regression
- Bayseian Linear Regression

Linear Regression

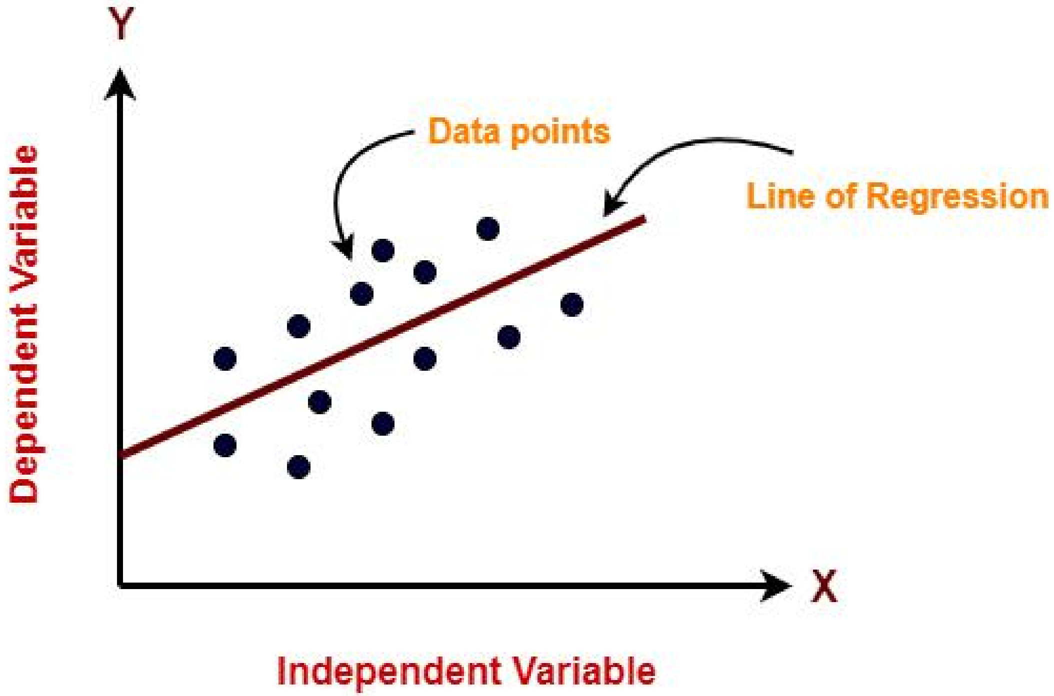
- Linear regression model represents the linear relationship between a dependent variable and independent variable(s) via a sloped straight line.
- y=mx+c

    y= dependent variable(continous)

    x= independent variable

    m= slope

    c=y intercept
- $m=(y_2-y_1)/(x_2-x_1)$
  - x_1 and y_1 coordinate of first data point.
  - x_2 and y_2 coordinate of second data point.



Types of Linear Regression

Types of Linear Regression

Simple Linear Regression

$y=mx+c$

Multiple Linear Regression

$y=m_1 x_1+m_2 x_2+m_3x_3+........+m_nx_n+c$

Simple Linear Regression

- Simple linear regression is a statistical method that allows us to summarize and study relationships between two continuous (quantitative) variables.
- One variable, denoted x, is regarded as the predictor, explanatory, or independent variable.
- The other variable, denoted y, is regarded as the response, outcome, or dependent variable.

Application of Simple Linear Regression

- used to predict the Economic growth of country
- used to pridict the future price of any product
- used to predict the score of players based on previous performance.
- house price pridiction

Error Matrix

There are three error metrics that are commonly used for evaluating and reporting the performance of a regression model.

They are:

- Mean Squared Error (MSE).
- Root Mean Squared Error (RMSE).
- Mean Absolute Error (MAE)

Error Matrix Formulas

MSE= $\frac{1}{n}\sum_{i=1}^{n}(y_i - \ y_{pre})^2$

RMSE= $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \ y_{pre})^2}$

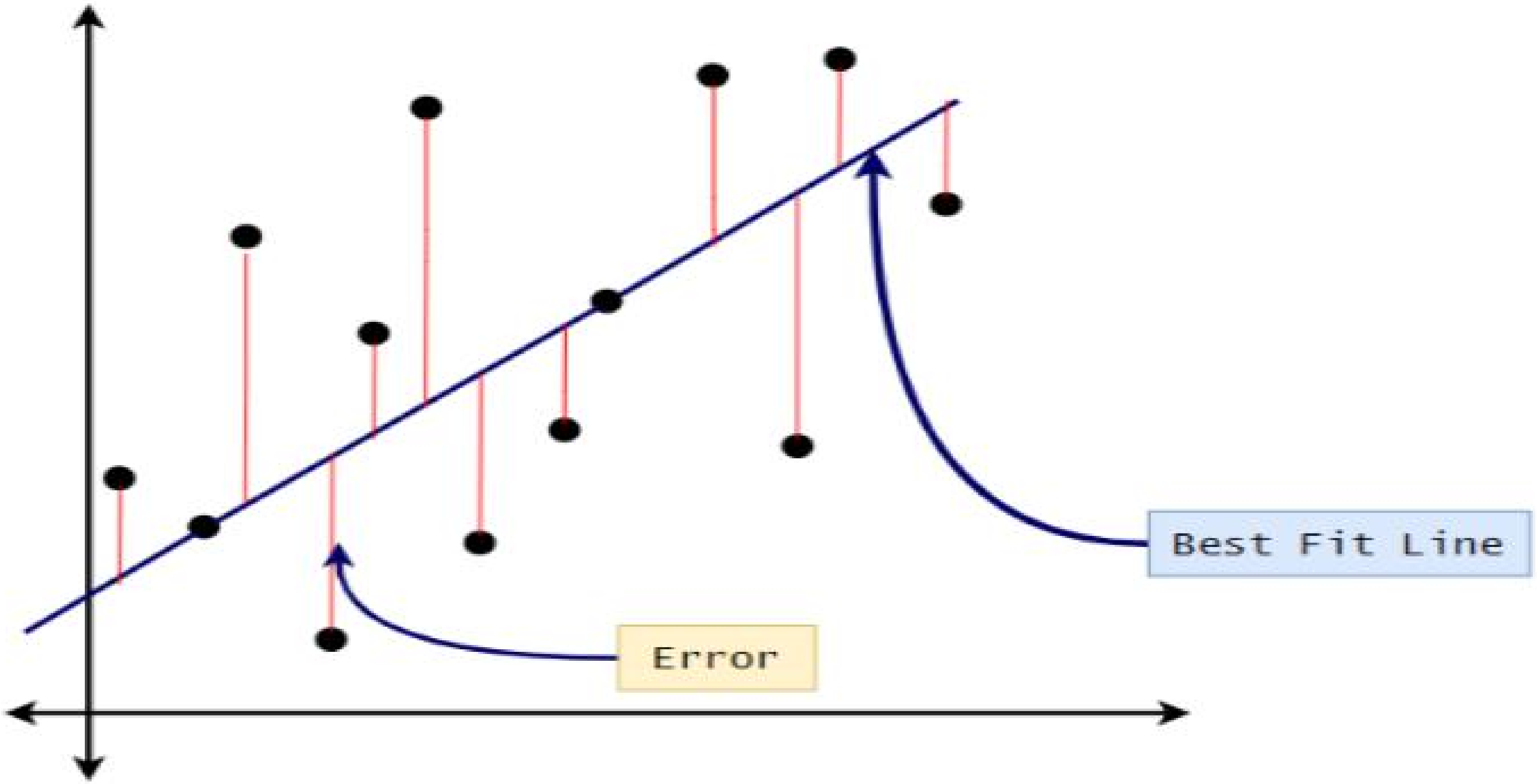
MAE= $\frac{1}{n}\sum_{i=1}^{n}\left|y_i - \ y_{pre}\right|$

Formulas

- y=mx+c
- cofficient of line $m=\frac{n\sum xy-\sum x\sum y}{n\sum x^2-(\sum x)^2}$
- y-intercept $c=\frac{\sum x^2\sum y-\sum x\sum xy}{n\sum x^2-(\sum x)^2}$
- mean square error $\text{MSE}=\frac{1}{n}\sum_{i=1}^{n}(y_i-y_{pre})^2$

where y_i is actual value and y_{pre} is predicted value

Mean Square Error



Accuracy of linear Regression

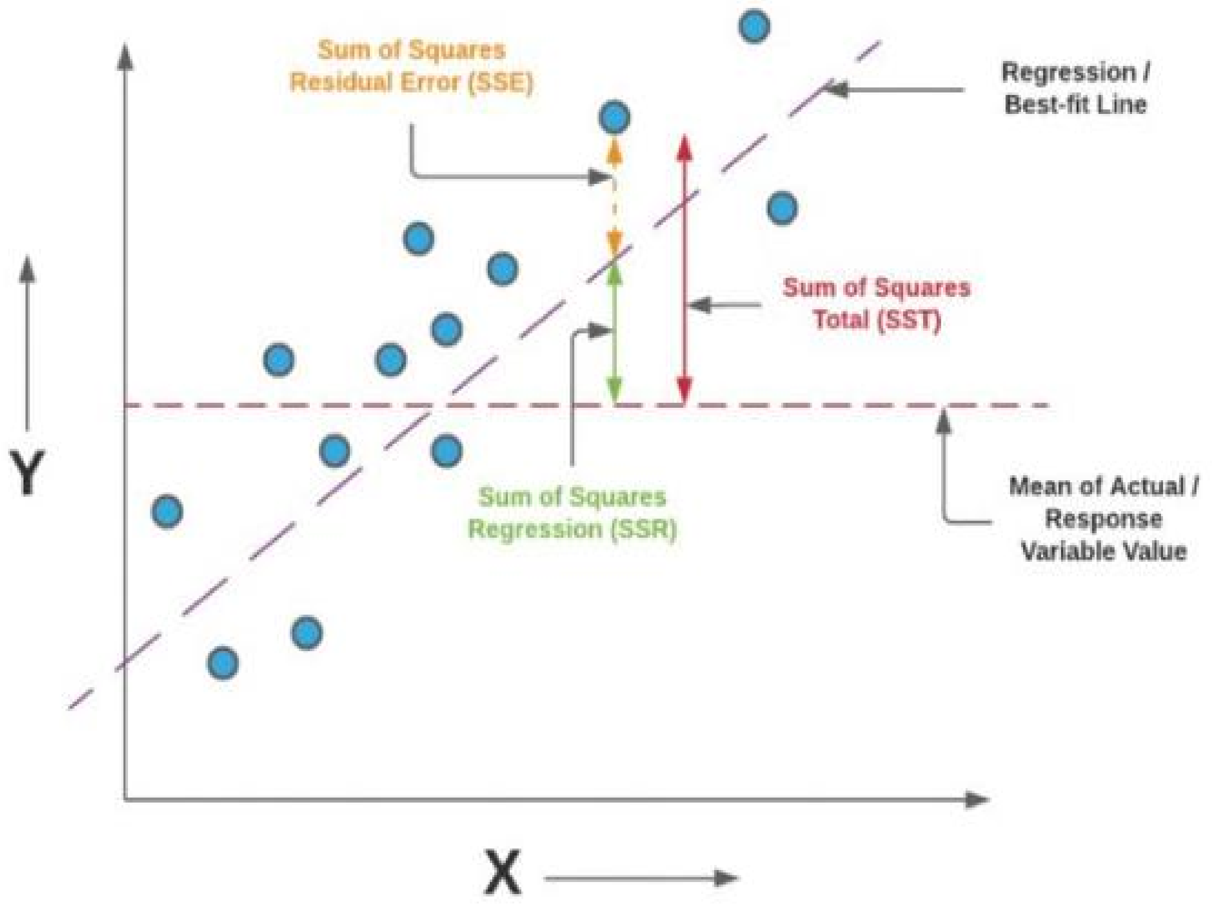
- Accuracy of linear Regression measured by
  - loss
  - R squared
  - Adjested R squared

R square

$$R^2 = \frac{SSR}{SST} * 100$$

$$SSR = \sum (y_{pre} - y_{mean})^2$$

SST= $\sum (y_i - y_{mean})^2$



Example 1:
predict salary based on experience

See excel file: Salary_data.csv